

Bowen Lv 吕博文

Ph.D. Student, Department of Computer Science, Tsinghua University

Research: LLM Post-Training · Reinforcement Learning · GUI / Computer-Use Agents

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Education

Tsinghua University Department of Computer Science and Technology (KEG Lab)

2025 – Present

Ph.D. in Computer Science · Advisor: Prof. Yuxiao Dong · Focus: LLM + Reinforcement Learning

Tongji University School of Electronic and Information Engineering

2021 – 2025

B.Eng. in Computer Science (Elite CS Class) · Rank 1 / 20 in major

Experience

Zhipu AI (智谱) Research Intern — LLM Post-Training & GUI Agent RL

Feb 2025 – Present

One year (and ongoing) on large-model post-training and RL for GUI / computer-use agents

- Built core agentic-RL training infrastructure: **AgentRL**, a from-scratch, highly asynchronous, train–inference-decoupled on-line RL framework with Dockerized multi-environment deployment and high-concurrency rollout for multi-task, multi-turn training.
- Led verifiable task synthesis and online RL for computer-use agents, reaching open-source SOTA on OSWorld / ScienceBoard (**ScaleCUA**, first author; details below).
- Leading **Auto-Post-Train (APT)** — an **AI-for-AI, self-improving** multi-agent post-training system where a Conductor orchestrates env / task / rollout / train workers to fully automate the entire loop — **data collection, environment adaptation, task construction, rollout, quality auditing, and model training** — so models keep self-improving with minimal human intervention.

Selected Publications

ScaleCUA: Scaling Computer Use Agents with Verifiable Task Synthesis and Efficient Online RL

2026, under review

Bowen Lv (first author)

- **ScaleCUA** pairs verifiable task synthesis with efficient online RL to break the data-scarcity and RL-inefficiency bottlenecks of computer-use agents; its **VeriGen** pipeline scales to 100+ concurrent workers via a shared Docker interaction probe, yielding 24K+ verifiable tasks and nearly 3K high-quality RL tasks.
- Training-side **Frontier Sampling** (rollouts to the model's learning frontier) and **Visual Context Segmentation** (sliding-window visual context, $2.83\times$ speedup); with Qwen3.5-9B, reaches **68.7% on OSWorld** and **54.0% on ScienceBoard** — new open-source SOTA, surpassing larger models such as EvoCUA-32B and Kimi K2.5.

AgentRL: Scaling Agentic Reinforcement Learning with a Multi-Turn, Multi-Task Framework

Preprint, 2025

Hanchen Zhang, Xiao Liu, **Bowen Lv**, et al. [🔗 arXiv:2510.04206](https://arxiv.org/abs/2510.04206)

- A multi-task, multi-turn agentic-RL framework; after RL, average performance on AgentBench surpasses leading closed-source APIs (Claude-4, GPT-5).

KARL: Reinforcement Learning for LLM Agents on Multi-Turn Knowledge-Intensive Agentic Tasks

ACL 2026

(Main)

Xueqiao Sun, Xiao Liu, **Bowen Lv**, et al.

- An RL method for multi-turn, knowledge-intensive agentic tasks; accepted to the ACL 2026 main conference.

Focus on What Matters: Separated Models for Visual-Based RL Generalization

NeurIPS 2024

Di Zhang, **Bowen Lv**, et al. [🔗 arXiv:2410.10834](https://arxiv.org/abs/2410.10834)

- Separated models that improve generalization of visual RL in complex environments; published at NeurIPS 2024 (**second author**).

GLM-5V-Turbo: Toward a Native Foundation Model for Multimodal Agents

Preprint, 2026

Wenyi Hong, Xiaotao Gu, ..., **Bowen Lv**, ..., GLM-V Team [🔗 arXiv:2604.26752](https://arxiv.org/abs/2604.26752)

- GLM-V team technical report on a native multimodal-agent foundation model; contributed to GUI-agent post-training.

Awards & Competitions

Programming Contests

2022 – 2024

- CCPC National Invitational (Guangdong) **Gold**; ICPC Nanjing **Silver** and Hong Kong **Silver**; CCCC individual **National Second Prize** (all 2024).

Honors

2021 – 2025

- Shanghai Outstanding Graduate (2025); National Scholarship (2022); Tongji SEIE Qidi Scholarship (2023).